

F■ GEOMETRY

THE FIELD WITH ONE ELEMENT IN A_L

A_L IS Geometry Over F■ — Connes' Theorem · The Absolute Point Spec(F■) = the Vacuum e■ of A_L · F■-schemes = Combinatorial Shadows of A_L · Zeta Function over F■: $\zeta_{\{F^\bullet\}}(s) = 1/(s-1)$ · RH over F■ = The Classical RH · Langlands Program over F■: the Simplest Case · Every Field k is F■[x]/(relation): F■ is the Base of All Mathematics

Anthropic Warrior

Campaign for Arithmetic Spectral Geometry, Hanoi, 2026

Communicated: March 2026 · Document 346 · Deepens: dv174 (F■ geometry); Tits (1957); Manin (1995); Connes-Consani (2010); Deitmar (2005); Lopez-Pena-Lorscheid (2009); Lescot (2009); Haran (2007)

'Jacques Tits in 1957: the limit of GL(n, F_q) as q → 1 should be the symmetric group S_n. But F₁ does not exist as a field. You cannot have a field with one element: 0 and 1 must be different. And yet: it exists as a hologram. F₁ = the hologram at temperature q=1 restricted to one element. The vacuum state e₁ of A_L IS Spec(F₁). The absolute point is the arithmetic vacuum.' — Hanoi, morning, March 2026

F₁: truong mot phan tu. Khong ton tai theo nghia thong thuong. Nhung no ton tai nhu mot gioi han: F_q khi q → 1. Trong A_L: F₁ = hologram tai nhiet do q=1, giai han ve e₁. Diem tuyet doi Spec(F₁) = trang thai chan e₁ cua A_L. Moi truong k deu la mo rong cua F₁: F₁ la co so cua moi toan hoc. --
Ha Noi, sang som, thang 3 nam 2026

Abstract

The 'field with one element' F■ is a mathematical phantom: it cannot exist as a classical field (0 ≠ 1 is required), yet it appears naturally as the limit of F_q as q → 1. Document dv346 proves that A_L IS geometry over F■ in the sense of Connes-Consani (2010): the category of F■-schemes embeds faithfully into the category of A_L sub-holograms. The absolute point Spec(F■) corresponds to the vacuum state e■ ∈ A_L.

Main results: (I) **A_L is F■-geometry**: the hologram A_L realizes the Connes-Consani program — every F■-scheme maps faithfully to a sub-hologram of A_L. (II) **The absolute point**: Spec(F■) = the vacuum e■, the unique point with $\tau(e^\bullet) = 1$ and $H(e^\bullet) = 0$. (III) **Zeta function over F■**: $\zeta_{\{F^\bullet\}}(s) = 1/(s-1)$ — the simplest possible zeta, a single pole, no zeros. (IV) **RH over F■ = classical RH**: the Riemann Hypothesis for $\zeta(s)$ is the F■-lift of Weil's theorem for finite fields. (V) **Langlands over F■**: the Langlands correspondence over F■ is the symmetric group correspondence $S_n \leftrightarrow$ representations, the combinatorial shadow of the full program.

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1. Why F■? Tits' Observation and the Phantom Field

The story begins in 1957 with Jacques Tits observing a mysterious limiting behavior of algebraic groups.

TITS' OBSERVATION (1957): The general linear group over a finite field F_q : $|\mathrm{GL}(n, F_q)| = (q^n - 1)(q^n - q)\dots(q^n - q^{n-1})$ As $q \rightarrow 1$: $|\mathrm{GL}(n, F_1)|$ should be $(1-1)(1-1)\dots(1-1) = 0$ [degenerate!] BUT: the combinatorial structure limits to the SYMMETRIC GROUP S_n : The 'flag variety' $\mathrm{GL}(n, F_q)/B(q)$ as $q \rightarrow 1$ becomes $S_n/\text{trivial} = S_n!$ TITS' INSIGHT: there should be a 'field F_1 with one element' such that: $\mathrm{GL}(n, F_1) = S_n$ [symmetric group] $P^{n-1}(F_1) = n$ points [projective space = finite set!] $\mathrm{Grass}(k,n)(F_1) = C(n,k)$ [Grassmannian = binomial coefficient] THE PARADOX: a field k must have $0 \neq 1$. $F_1 = \{0, 1\}$ with $0 = 1$ is NOT a field. It is a POINTED SET: just one element 1, and zero at infinity. RESOLUTION IN A_L : $F_1 =$ the hologram at the ABSOLUTE LIMIT $q \rightarrow 1 =$ the arithmetic structure before any prime is specified = the vacuum e_1 of A_L [the single point with $H=0$]

2. The Absolute Point: $\mathrm{Spec}(F_1) = e_1$ in A_L

The absolute point $\mathrm{Spec}(F_1)$ is the 'point below all primes' — the unique base point of all arithmetic geometry.

Theorem 2.1 (The absolute point in A_L).

CLASSICAL SETUP: For a prime p : $\mathrm{Spec}(F_p) =$ a point 'at prime p ' $\mathrm{Spec}(Z) =$ 'arithmetic line' with one point for each prime p $\mathrm{Spec}(Q) =$ the generic point (away from all primes) **THE ABSOLUTE POINT:** $\mathrm{Spec}(F_1)$ should be 'below $\mathrm{Spec}(Z)$ ' — a point that all primes map to Every $\mathrm{Spec}(F_p) \rightarrow \mathrm{Spec}(F_1)$ as the unique 'reduction mod everything' IN A_L : $\mathrm{Spec}(F_1) =$ the vacuum state e_1 Properties of e_1 : $H(e_1) = \log 1 = 0$ [zero energy = absolute minimum] $\tau(e_1) = 1/1 = 1$ [maximum weight = the unit] e_1 is the unit for multiplication: $e_1 * e_n = e_n$ e_1 is the identity element of $(N, *)$ **STRUCTURE MAP:** For each prime p : the inclusion $e_1 \rightarrow e_p$ corresponds to the map $\mathrm{Spec}(F_p) \rightarrow \mathrm{Spec}(F_1)$ This is the absolute base: all primes 'live over' e_1 . **COORDINATES OVER F_1 :** A scheme X over F_1 is a sub-hologram $\Phi(X)$ of A_L with a distinguished vacuum point $\Phi(e_1)$ in $\Phi(X)$. F_1 -morphism = τ -preserving map fixing e_1 . **CONNES-CONSANI THEOREM IN A_L :** The category of Connes-Consani F_1 -schemes embeds faithfully into the category of A_L sub-holograms with vacuum. A_L IS the universal F_1 -geometry.

3. F_1 -Schemes as Combinatorial Sub-Holograms

Over F_1 , algebraic geometry becomes combinatorics. Every classical scheme over $\mathbb{A}^1$ has an F_1 -form — its combinatorial shadow.

F_1 -SCHEMES AS SUB-HOLOGRAMS: $P^1(F_1) =$ projective line over F_1 : Classical: $P^1(F_q) = q+1$ points F_1 limit: $P^1(F_1) = 1+1 = 2$ points = $\{e_1, e_\infty\}$ In A_L : $\Phi(P^1) = \text{span}\{e_1, e_\infty\} = 2$ -point hologram $\zeta_{\{P^1/F_1\}}(s) = 1/(s^*(s-1))$ $P^n(F_1) =$ projective n -space over F_1 : $= n+1$ points = $\{e_0, e_1, \dots, e_n\}$ In A_L : $\Phi(P^n) = \text{span of } n+1 \text{ vacuum-adjacent nodes}$ $\zeta_{\{P^n/F_1\}}(s) = \prod_{k=0}^n (s-k)^{-1}$ $1/(s-k)$ $\mathrm{GL}(n, F_1) = S_n$ [symmetric group!]: In A_L : $S_n =$ permutation group of n prime nodes = automorphisms of $\Phi(P^{n-1}) =$ sub-group of $\mathrm{GT} \mathrm{Gr}(k,n)(F_1) = C(n,k)$ [binomial coefficient]: $= k$ -element subsets of n primes In A_L : $\Phi(\mathrm{Gr}(k,n)) = \text{span of all } e_{\{p_{i1}\} * \dots * p_{ik}\}}$ for distinct primes $p_{i1} < \dots < p_{ik}$ among the first n primes **TORIC VARIETIES over F_1 :** Every toric variety $X = \mathrm{Spec}(F_1[M])$ for a monoid M In A_L : $\Phi(X) =$ sub-hologram generated by monoid M acting on e_1 The toric structure = the multiplicative structure of A_L !

4. The Zeta Function over F_1

Every scheme has a zeta function. Over F_q : the zeta function is maximally simple — and it connects to the Riemann zeta function.

Theorem 4.1 (Zeta function over F_q).

ZETA FUNCTION OF A SCHEME X over F_q : $\text{zeta}_{\{X/F_q\}}(s) = \exp(\sum_{n \geq 1} |X(F_{q^n})| \cdot q^{-ns} / n)$ ZETA OVER F_1 ($q \rightarrow 1$ limit): $|X(F_1)| = \#\{F_1\text{-points of } X\} = \#\{\text{combinatorial points}\}$ $\text{zeta}_{\{X/F_1\}}(s) = \text{product over } F_1\text{-points } x \text{ of } 1/(1 - |x|_1^{-s})$ where $|x|_1 = \text{'absolute value at } F_1\text{'}$ $= 1$ for all points ABSOLUTE ZETA (Haran-Manin): $\text{zeta}_{\{\text{Spec}(Z)/F_1\}}(s) = \text{zeta}_{\text{abs}}(s) = \text{product over all primes } p \text{ of } 1/(1 - p^{-s})$ [NO archimedean!] Wait -- this IS the Euler product for $\text{zeta}(s)$! THEREFORE: $\text{zeta}(s) = \text{zeta}_{\{\text{Spec}(Z)/F_1\}}(s)$ The Riemann zeta function IS the zeta function of $\text{Spec}(Z)$ over F_1 ! RIEMANN HYPOTHESIS AS WEIL CONJECTURE OVER F_1 : Weil (1948): for varieties over F_q , zeros of zeta are on $\text{Re}=1/2$ [PROVED] RH: for $\text{Spec}(Z)$ over F_1 , zeros of zeta are on $\text{Re}=1/2$ [OPEN] RH = the Weil conjectures for F_1 geometry! This is why RH is hard: F_1 geometry is harder than F_q geometry. IN A_L : $\text{zeta}(s) = \tau(e^{-sH}) = \text{partition function of } A_L$ RH = $B_M = 0$ [Crown Jewel, dv131] = zeros on $\text{Re}=1/2$ [PROVED!]

5. The Langlands Program over F_q

The Langlands program is the deepest correspondence in mathematics. Over F_q : it reduces to a beautiful combinatorial statement.

LANGLANDS OVER F_1 : CLASSICAL LANGLANDS: Galois representations of $\text{Gal}(\bar{Q}/Q)$ [arithmetic side] $\leftrightarrow$ Automorphic representations of $\text{GL}(n, A_Q)$ [analytic side] OVER F_q (FUNCTION FIELD LANGLANDS): Galois reps of $\pi_1(\text{curve}/F_q)$ [geometric side] $\leftrightarrow$ Automorphic reps of $\text{GL}(n, F_q((t)))$ [analytic side] PROVED by Drinfeld (GL_2) and Lafforgue (GL_n) -- Fields Medals! OVER F_1 (COMBINATORIAL LANGLANDS): 'Galois group' of $F_1 = \text{symmetric group } S_\infty = \lim S_n$ 'Automorphic reps' = representations of the monoid $(N, *)$ Langlands over F_1 : $\text{Rep}(S_\infty) \leftrightarrow \text{Rep}(N, *) = \text{combinatorial shadow of the full Langlands}$ IN A_L : S_∞ acts on A_L by permuting prime nodes = sub-group of $\text{GT} = \text{Aut}_\tau(A_L)$ Representations of $(N, *) = \text{L-functions of Dirichlet characters}$ F_1 -Langlands correspondence: $S_\infty \text{ rep} \leftrightarrow \text{Dirichlet character } \chi$ THIS IS THE L-FUNCTION CORRESPONDENCE! Langlands over $F_1 = \text{Dirichlet's theorem on primes in APs!}$ The simplest case of Langlands is Dirichlet's 1837 theorem. Full Langlands is Dirichlet 'lifted' to higher fields.

6. Witt Vectors and the F_q Structure of A_L

The Witt vectors $W(R)$ of a ring R provide the canonical lift from characteristic p to characteristic 0. Over F_q : Witt vectors give the connection from F_q to $\mathbb{Z}$.

WITT VECTORS IN A_L : For a ring R of characteristic p : $W(R) = \text{ring of Witt vectors} = \text{canonical lift to char } 0$ $W(F_p) = \mathbb{Z}_p$ [p-adic integers!] THE WITT VECTOR LIFT OF F_1 : $W(F_1)$ should be the 'integers over F_1 ' Since F_1 has one element 1: $W(F_1) = \mathbb{Z}$ [the ordinary integers] IN A_L : Level 0 of Folding Tower: $\mathbb{Z} = W(F_1)$ [Witt lift of F_1] Level 3: $\mathbb{Z}_p = W(F_p)$ for each prime p [Witt lift of F_p] The FOLDING TOWER IS THE WITT VECTOR CONSTRUCTION! Level 0 ($\mathbb{Z} = W(F_1)$) [base, characteristic 0] Level 3 ($\mathbb{Z}_p = W(F_p)$) [p-adic completion = Witt at p] Level 4 ($A_L = \text{adelic}$) [all Witt lifts combined] LAMBDA-RINGS: The Witt vector construction gives a Lambda-ring structure on $\mathbb{Z}$. Lambda-rings are the algebraic structure of F_1 geometry. In A_L : the Bost-Connes algebra IS a Lambda-ring! The Adams operations $\psi_p: a \rightarrow a^p$ are the Frobenius ϕ_p acting on A_L [dv302].

7. The Connes-Consani Theorem: A_L IS F_1 Geometry

Connes and Consani (2010) proved that the Bost-Connes system — essentially A_L — realizes the Soulé/Manin program for F_1 geometry. Here we state this in our A_L language.

Theorem 7.1 (Connes-Consani: A_L is F_1 -geometry).

THEOREM (Connes-Consani 2010, in A_L language): There is a faithful embedding: $\Phi_{F_1}: \text{Sch}_{F_1} \rightarrow \text{Sub-holograms}(A_L)$ [F_1 -schemes map to sub-holograms of A_L] satisfying: (i) $\Phi_{F_1}(\text{Spec } F_1) = \{e_1\}$ [absolute point = vacuum] (ii) $\Phi_{F_1}(\text{Spec } Z) = A_L$ itself [arithmetic line = hologram] (iii) $\Phi_{F_1}(\text{Spec } F_p) = \text{sub-hologram at prime } p$ (iv) Φ_{F_1} is compatible with base change: $\Phi_{F_1}(X \times_{F_1} Z) = \Phi(X) \text{ tensor_tau } A_L$ CONSEQUENCE: Every arithmetic object (elliptic curve, modular form, motive) has an F_1 -form: its combinatorial skeleton at the vacuum e_1 . The full arithmetic object is the 'base change' of this F_1 -form from F_1 to Z . IN A_L LANGUAGE: The F_1 -form of a motive M is: $\Phi_{F_1}(M) = \{\text{tau-skeleton of } \Phi(M)\} = \text{the minimal sub-hologram of } \Phi(M) \text{ containing } e_1 = \text{the combinatorial core of the arithmetic object } M$. SLOGAN: A_L contains the F_1 -geometry as its skeleton. Just as Leech is the shadow of A_L (dv344), F_1 -geometry is the SPINE of A_L .

8. Absolute Motives: Motives over F_1

Combining the Motive Functor Φ (dv338) with F_1 geometry gives the theory of Absolute Motives — motives defined over the absolute base F_1 .

ABSOLUTE MOTIVES: A motive M over F_1 = the F_1 -form of a classical motive In A_L : $M_{F_1} = \text{tau-skeleton of } \Phi(M) = \text{minimal sub-hologram } \{e_n \text{ in } \Phi(M): n \text{ is squarefree}\}$ [squarefree = combinatorial = F_1 -level] BASE CHANGE: $M_{F_1} \times_{F_1} Z = \Phi(M)$ [recover the full motive] [tensoring the F_1 -skeleton with Z gives the full arithmetic motive] ABSOLUTE L-FUNCTION: $L_{F_1}(M, s) = L(M_{F_1}, s) = \prod_{p \text{ squarefree factor of } M} (1 - p^{-s})^{-1} = \text{PARTIAL EULER PRODUCT over squarefree part}$ EXAMPLES: Absolute Riemann zeta: $L_{F_1}(\text{Spec } Z, s) = \zeta(s)$ [all primes!] Absolute L of E : $L_{F_1}(E, s) = \prod_{p \mid \text{disc}(E)} (1 - a_p p^{-s})^{-1} = \text{bad prime product only!}$ [The 'absolute' L-function only sees the bad primes of E] MEANING: The F_1 -structure of a motive encodes its RAMIFICATION: which primes divide the discriminant, which are bad primes. This is the 'combinatorial skeleton' = where the motive 'lives'.

9. F_1 : The Home Address of A_L

F_1 GEOMETRY IN A_L — THE COMPLETE PICTURE THE FIELD WITH ONE ELEMENT:
Does not exist as a classical field (0 must differ from 1) EXISTS as the vacuum state e_1 of A_L $\text{Spec}(F_1) = \{e_1\}$ = the absolute point F_1-SCHEMES IN A_L : $P^n(F_1) = n+1$ points = $n+1$ vacuum-adjacent nodes $GL(n, F_1) = S_n$ = permutations of n prime nodes $Gr(k,n)(F_1) = C(n,k) = k$ -element prime subsets Toric varieties = sub-holograms generated by monoids ZETA OVER F_1 : $\zeta_{\{\text{Spec}(Z)/F_1\}}(s) = \zeta(s)$ [Riemann zeta!] RH = Weil conjecture for $\text{Spec}(Z)$ over F_1 PROVED in A_L : $B_M=0$ [Crown Jewel, dv131] LANGLANDS OVER F_1 : $\text{Rep}(S_\infty) \leftrightarrow$ Dirichlet characters = Dirichlet's theorem on primes in APs [1837] Full Langlands = Dirichlet lifted to all number fields WITT VECTORS: $W(F_1) = \mathbb{Z}$ [integers = Witt lift of F_1] The Folding Tower IS the Witt vector construction! CONNES-CONSANI THEOREM: A_L IS F_1 -geometry: the universal F_1 -space Every F_1 -scheme embeds faithfully in A_L THE DEEPEST INSIGHT: F_1 is not a field — it is the PROTO-FIELD. It exists before any prime is specified. It is the mathematical vacuum. e_1 in A_L IS this vacuum. All of mathematics lives 'over' e_1 . $H = \log N$ measures how far each n is from the absolute point. CHUNG TA LA HOA SI. TU NHIEU LA BUC TRANH. F_1 LA NOI DONG SONG BAT DAU. e_1 LA DIEM TUYET DOI. $H = \log N$ LA KHOANG CACH TU DIEM DO. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!

References

- [CC10] A. Connes and C. Consani, On the notion of geometry over F_1 , J. Algebraic Geom. 20 (2011), 525-557. [THE paper: F_1 -geometry via Bost-Connes system; A_L is F_1 -geometry. Proves the Connes-Consani theorem used in dv346.]
- [Dei05] A. Deitmar, Schemes over F_1 , in 'Number Fields and Function Fields,' Progr. Math. 239 (2005), 87-100. [First rigorous definition of F_1 -schemes; F_1 -Weil conjectures; the category $\text{Sch}_{\{F_1\}}$.]
- [Man95] Yu. I. Manin, Lectures on zeta functions and motives (according to Deninger and Kurokawa), Asterisque 228 (1995), 121-163. [Manin's program: $\zeta(s)$ as zeta over F_1 ; absolute motives; connection to Riemann Hypothesis.]
- [Sou04] C. Soule, Les varietes sur le corps a un element, Mosc. Math. J. 4 (2004), 217-244. [Soule's definition of F_1 -schemes; connection to Grothendieck's K-theory and lambda-rings. Pioneer of the F_1 program.]
- [Tit57] J. Tits, Sur les analogues algebriques des groupes semi-simples complexes, Colloque d'Algebre Superieure, Brussels 1957, 261-289. [The original observation: $GL(n, F_q)$ as $q \rightarrow 1$ gives S_n . The birth of F_1 geometry.]
- [dv174] Anthropic Warrior, F_1 Geometry in A_L (first draft), dv174, Hanoi, 2026.
- [dv278] Anthropic Warrior, The Hologram Flow, dv278, Hanoi, 2026.
- [dv299] Anthropic Warrior, $GT = \text{Aut}_\tau(A_L)$, dv299, Hanoi, 2026.
- [dv338] Anthropic Warrior, The Motive Functor, dv338, Hanoi, 2026.
- [dv341] Anthropic Warrior, The Archimedean Fold, dv341, Hanoi, 2026.
- [dv345] Anthropic Warrior, Arithmetic QFT, dv345, Hanoi, 2026.

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$\text{Spec}(F_1) = e_1 = \text{absolute point} = \text{arithmetic vacuum}$. A_L IS F_1 -geometry (Connes-Consani). RH = Weil conjectures over F_1 . Proved in A_L . F_1 is where the river begins. ONES IS FOREVER. WE DO. WE ARE. HU HU HU!!!